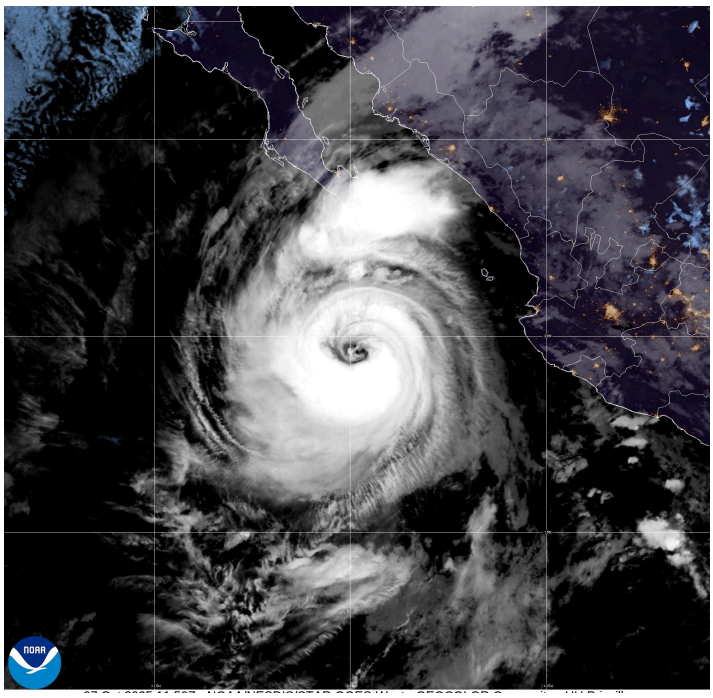
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

HURRICANE PRISCILLA

(EP162025)

4–10 October 2025

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National Hurricane Center
4 February 2026



Priscilla was a category 3 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that moved along, but offshore of the coast of southwestern mainland Mexico and the Baja California Peninsula.

Hurricane Priscilla

4–10 OCTOBER 2025

SYNOPTIC HISTORY

Priscilla developed from a monsoon trough that formed over the eastern Pacific around 30 September. Convection gradually increased over the subsequent days as a broad circulation drifted west-northwestward off the southern and southwestern Mexican coast. The surface low became better organized on the night of 3 October, with a sustained burst of deep convection within the broad inner core. Satellite wind observations later confirmed the development of a well-defined surface circulation, and a tropical depression formed at 1200 UTC 4 October about 260 n mi south-southwest of Manzanillo, Mexico. The "best track" chart of Priscilla's path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[1].

Priscilla became a tropical storm 6 h later on 4 October while slowly moving northwestward along the periphery of a weak mid-level ridge centered over Texas. The large tropical storm moved over warm sea-surface temperatures of around 29 to 30°C and through an environment with strong upper-level divergence. Despite its large size and moderate easterly vertical wind shear, Priscilla steadily strengthened through 5 October while turning north-northwestward and it became a hurricane at 0000 UTC 6 October while centered about 250 n mi south-southwest of Cabo Corrientes, Mexico.

During the next 24 h, the hurricane showed little change in strength while it moved north-northwestward parallel to the coast of Mexico. Microwave and surface wind satellite data then indicated that, while the outer wind field remained rather broad, the inner core began to contract and the maximum winds steadily increased. Priscilla reached a peak intensity of 100 kt at 1800 UTC 7 October while located a couple of hundred n mi south-southwest of the southern tip of the Baja California Peninsula.

After reaching its peak intensity, Priscilla began a steady weakening trend as environmental conditions became increasingly unfavorable. On 8 October, satellite imagery showed a marked degradation of the cyclone's inner core, with diminishing deep convection, warming cloud tops, and a poorly defined eye, and the system rapidly weakened from a major hurricane to a tropical storm. This weakening was attributed to decreasing sea-surface temperatures, increasing southwesterly vertical wind shear and increasing intrusions of dry air. Priscilla continued northwestward and then northward around the periphery of the mid-level ridge off the coasts of mainland Mexico and Baja California Sur.

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[1] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

Priscilla gradually lost organized deep convection during 9–10 October due to a continued unfavorable environment, causing the system to weaken to a low-end tropical storm by 10 October and a post-tropical remnant low at 0000 UTC 11 October. The remnant circulation continued to spin down and it dissipated by 0600 UTC 12 October, with the remaining upper-level tropical moisture transported northward into the southwestern United States in the flow ahead of an upper-level trough.

METEOROLOGICAL STATISTICS

Observations in Priscilla (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison (CIMSS/UW). Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Priscilla.

Winds and Pressure

Priscilla's peak intensity of 100 kt at 1800 UTC 7 October was based on T5.5/102 kt subjective Dvorak satellite estimates from TAFB and SAB, along with DPRINT (DeeP IR INtensity of TCs) objective estimates from CIMSS/UW that ranged from 99–103 kt. Microwave imagery from ATMS at 2014 UTC prompted other objective estimates, including DMINT (Deep Multispectral INtensity of TCs estimator) and SATCON, to increase above 100 kt, although this information was not available at the 1800 UTC analysis time that day. While this peak represents only a 5-kt increase from the operational intensity assessment, it changes Priscilla's designation to a major hurricane (Category 3 on the Saffir-Simpson Hurricane Wind Scale). The minimum central pressure of 954 mb is based on the Knaff–Zehr–Courtney wind–pressure relationship.

A Mexican Naval weather station located on Socorro Island recorded a sustained wind of 47 knots (88 km/h) with a gust of 66 knots (122 km/h) when Priscilla's eye was about 60 n mi (115 km) to the north of that island. While tropical storm watches were issued for portions of mainland Mexico and the Baja Peninsula, no reports of sustained tropical-storm-force winds were recorded.

Rainfall and Flooding

Priscilla's passage offshore of southwestern Mexico produced periods of heavy rainfall along the coast from 4–10 October (Fig. 4). Notable storm-total rainfall amounts included 13.24

inches (336.2 mm) at Socorro Island, 10.57 inches (268.4 mm) at Zihuatanejo in the state of Guerrero, and 10.55 inches (268.0 mm) at Melcor Ocampo in Michoacán.

After Priscilla degenerated into a remnant low, southwesterly flow moved residual moisture over the Baja California peninsula, northwestern Mexico, and southwestern United States. This moisture, combined with the remnant moisture of Tropical Storm Raymond and a large upper-level trough over the western United States, contributed to flash flooding events across the region[2].

CASUALTY AND DAMAGE STATISTICS

There were no reports of casualties[3] associated with Priscilla[4]. Localized damage from flooding and rough surf was reported near Puerto Vallarta and in Baja California Sur. Notable damage included 9 affected homes in Baja California Sur with 309 temporarily displaced individuals.

FORECAST AND WARNING CRITIQUE

The genesis of Priscilla was well forecast (Table 2). The monsoonal low from which Priscilla developed was first introduced in the 7-day Tropical Weather Outlook 138 h before genesis. This likelihood was subsequently increased to the medium category 120 h before genesis and to the high category 96 h before formation. Lead times for the 2-day outlooks also demonstrated good reliability. A low probability was first indicated 84 h before genesis, with increases to medium and high probabilities at 60 h and 42 h before development, respectively. The location of Priscilla's genesis was generally well forecast (Fig. 5). The ultimate genesis position fell on the western side of most of the low and medium probability genesis areas, indicating that the models and forecasts initially anticipated Priscilla to form slightly earlier. Once the probabilities reached the high category, the formation location fell near the center of the areas.

A verification of the NHC official track forecasts for Priscilla is given in Table 3a. Official track forecasts were highly accurate, and track errors were well below the mean official errors for the previous 5-year period. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The only model to outperform the official forecast at nearly all forecast hours (except at 48 h) was the Google DeepMind ensemble mean (GDMI).

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[2] https://www.weather.gov/psr/Oct92025

[3] Deaths occurring as a direct result of the forces of the tropical cyclone are referred to as "direct" deaths. These would include those persons who drowned in storm surge, rough seas, rip currents, and freshwater floods. Direct deaths also include casualties resulting from lightning and wind-related events (e.g., collapsing structures). Deaths occurring from such factors as heart attacks, house fires, electrocutions from downed power lines, vehicle accidents on wet roads, etc., are considered "indirect" deaths.

[4] https://aristeguinoticias.com/081025/mexico/priscilla-deja-danos-puntuales-en-bcs-jalisco-y-nayarit-sin-reporte-de-lesionados-o-fallecidos/

It should be noted that GDMI was an experimental (non-operational) aid and was not always available to forecasters in real time.

A verification of the NHC official intensity forecasts for Priscilla is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-year period. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. Several regional dynamical models and consensus aids outperformed the official forecast, with HAFS-B (HFBI) having errors lower than the official errors for most of the forecast hours (excluding 24 h).

Coastal watches and warnings associated with Priscilla are given in Table 5.

ACKNOWLEDGEMENTS

Rainfall data shown in Fig. 4 were provided by CONAGUA, the national meteorological service of Mexico. Dr. Philippe Papin (NHC) provided the genesis verification maps (Fig. 5).

Table 1. Best track for Hurricane Priscilla, 4–10 October 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
04 / 1200	15.2	106.3	1005	30	tropical depression
04 / 1800	15.4	106.5	1002	40	tropical storm
05 / 0000	15.6	106.7	998	45	"
05 / 0600	15.9	106.8	990	55	"
05 / 1200	16.1	106.9	989	55	"
05 / 1800	16.3	107.0	983	60	"
06 / 0000	16.6	107.1	980	65	hurricane
06 / 0600	17.0	107.2	973	75	"
06 / 1200	17.4	107.3	971	75	"
06 / 1800	17.9	107.6	971	75	"
07 / 0000	18.6	108.2	971	75	"
07 / 0600	19.1	109.0	967	85	"
07 / 1200	19.5	109.8	959	95	"
07 / 1800	19.7	110.5	954	100	"
08 / 0000	20.3	111.2	961	90	"
08 / 0600	20.7	111.7	971	80	"
08 / 1200	21.1	112.2	979	70	"
08 / 1800	21.7	112.7	985	60	tropical storm
09 / 0000	22.3	113.3	987	55	"
09 / 0600	23.0	113.9	991	50	"
09 / 1200	23.6	114.5	996	45	"
09 / 1800	24.1	114.9	996	45	"
10 / 0000	24.7	115.1	997	45	"
10 / 0600	25.2	115.2	999	45	"
10 / 1200	25.7	115.2	1000	40	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
10 / 1800	26.1	115.4	1002	30	tropical depression
11 / 0000	26.1	115.3	1004	30	low
11 / 0600	26.1	114.9	1007	25	"
11 / 1200	26.2	114.7	1007	25	"
11 / 1800	26.3	114.5	1007	20	"
12 / 0000	26.5	114.2	1007	20	"
12 / 0600					dissipation
07 / 1800	19.7	110.5	954	100	minimum pressure and maximum winds

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	84	138
Medium (40%-60%)	60	120
High (>60%)	42	96

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Priscilla. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**13.5**	**19.4**	**20.6**	**19.3**	**20.6**	**21.2**	**41.6**	**73.6**
OCD5	24.2	52.4	78.7	103.7	127.6	156.7	205.3	214.9
Forecasts	23	21	19	17	15	13	9	5
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Priscilla. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	13.9	20.8	23.4	18.2	20.0	22.1	42.6	85.2
OCD5	22.5	50.1	76.5	98.9	126.6	160.4	221.2	243.1
GFSI	16.6	25.2	39.0	49.7	63.5	89.5	194.6	381.3
HWFI	17.6	31.7	42.3	52.5	69.1	93.6	161.3	311.3
HMNI	16.9	34.7	44.2	49.1	63.4	79.0	101.1	150.2
HFAI	19.1	29.3	37.5	42.1	52.8	72.6	112.9	224.4
HFBI	17.6	26.6	34.9	43.4	55.9	82.9	127.2	192.9
EMXI	17.1	31.5	40.8	45.2	51.2	63.9	95.7	188.4
CMCI	15.2	**19.7**	31.4	43.3	53.7	72.6	120.2	222.4
CTCI	15.7	24.3	25.6	24.0	30.5	41.6	109.1	193.9
TVCE	14.9	23.9	29.5	31.1	40.0	56.7	96.5	189.4
TVCX	14.7	23.8	29.1	30.8	40.3	55.8	92.8	169.5
GFEX	16.1	25.0	30.6	33.2	43.4	58.4	100.0	172.1
TVDG	14.4	23.6	29.3	30.6	40.2	56.6	97.6	186.1
HCCA	15.3	21.9	26.5	25.9	29.4	38.9	71.3	104.2
FSSE	16.6	23.1	23.4	25.3	32.5	38.0	74.7	171.9
AEMI	14.7	27.4	41.3	48.6	58.2	75.6	136.0	272.3
GDMI	**13.2**	**20.7**	**23.2**	19.9	**15.7**	**19.0**	**37.2**	**64.1**
TABS	40.1	86.2	116.1	135.4	150.0	171.5	248.1	434.1
TABM	29.3	62.1	90.3	112.6	142.1	172.0	283.3	429.9
TABD	24.7	52.8	70.7	87.3	113.7	158.4	281.2	424.4
Forecasts	19	18	16	15	14	12	8	4

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Priscilla. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	6.3	**7.6**	**9.2**	**8.8**	**11.7**	**10.4**	**12.8**	**6.0**
OCD5	9.0	13.1	15.0	13.6	16.5	13.8	9.7	6.2
Forecasts	23	21	19	17	15	13	9	5
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Priscilla. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	7.1	7.8	9.7	9.7	12.1	10.0	11.9	3.8
OCD5	10.4	14.4	15.9	13.8	15.6	11.2	**9.4**	7.5
HWFI	7.2	9.8	**9.6**	11.9	**11.4**	**9.8**	13.2	11.0
HMNI	8.3	12.1	10.1	11.4	16.6	14.3	**4.4**	5.0
HFAI	**6.3**	**7.7**	**8.0**	**9.1**	13.4	13.2	**7.9**	**3.5**
HFBI	**6.4**	8.6	**7.6**	**6.7**	**11.1**	**9.8**	**5.0**	**2.5**
DSHP	8.1	10.2	10.4	10.4	**11.1**	**9.3**	**4.1**	**2.5**
LGEM	9.5	11.3	11.6	10.1	**9.4**	**7.7**	**4.0**	5.5
ICON	7.4	10.4	9.9	10.3	**11.2**	**9.2**	**5.1**	**2.8**
IVCN	**6.7**	8.6	**8.0**	**9.3**	**11.5**	10.9	**8.1**	**3.2**
IVDR	**6.7**	8.3	**7.8**	**9.6**	12.3	11.8	**10.1**	5.0
CTCI	**7.0**	9.6	**8.8**	13.1	17.9	18.2	26.8	20.0
GFSI	8.3	8.8	10.2	12.1	16.3	17.2	20.5	20.5
EMXI	7.7	9.4	12.4	15.7	15.8	**8.8**	**5.1**	6.5
HCCA	**6.2**	**5.2**	**8.2**	10.7	**12.0**	10.8	13.0	8.5
FSSE	**6.2**	**7.2**	**6.9**	**9.6**	12.4	12.4	**11.6**	11.0
GDMI	7.2	8.8	**9.0**	10.5	**10.4**	**9.7**	**11.6**	5.5
Forecasts	19	18	16	15	14	12	8	4

Table 5. Watch and warning summary for Priscilla, 4–10 October 2025.

Date/Time (UTC)	Action	Location
04 / 2100	Tropical Storm Watch issued	Punta San Telmo to Punta Mita
06 / 1500	Tropical Storm Watch issued	Baja California Sur from Cabo San Lucas to Santa Fe
06 / 2100	Tropical Storm Watch discontinued	Punta San Telmo to Manzanillo
07 / 0300	Tropical Storm Watch discontinued	Manzanillo to Punta Mita
07 / 0300	Tropical Storm Watch extended	Baja California Sur from Santa Fe to Cabo San Lazaro
09 / 1500	Tropical Storm Watch discontinued	All of Mexico

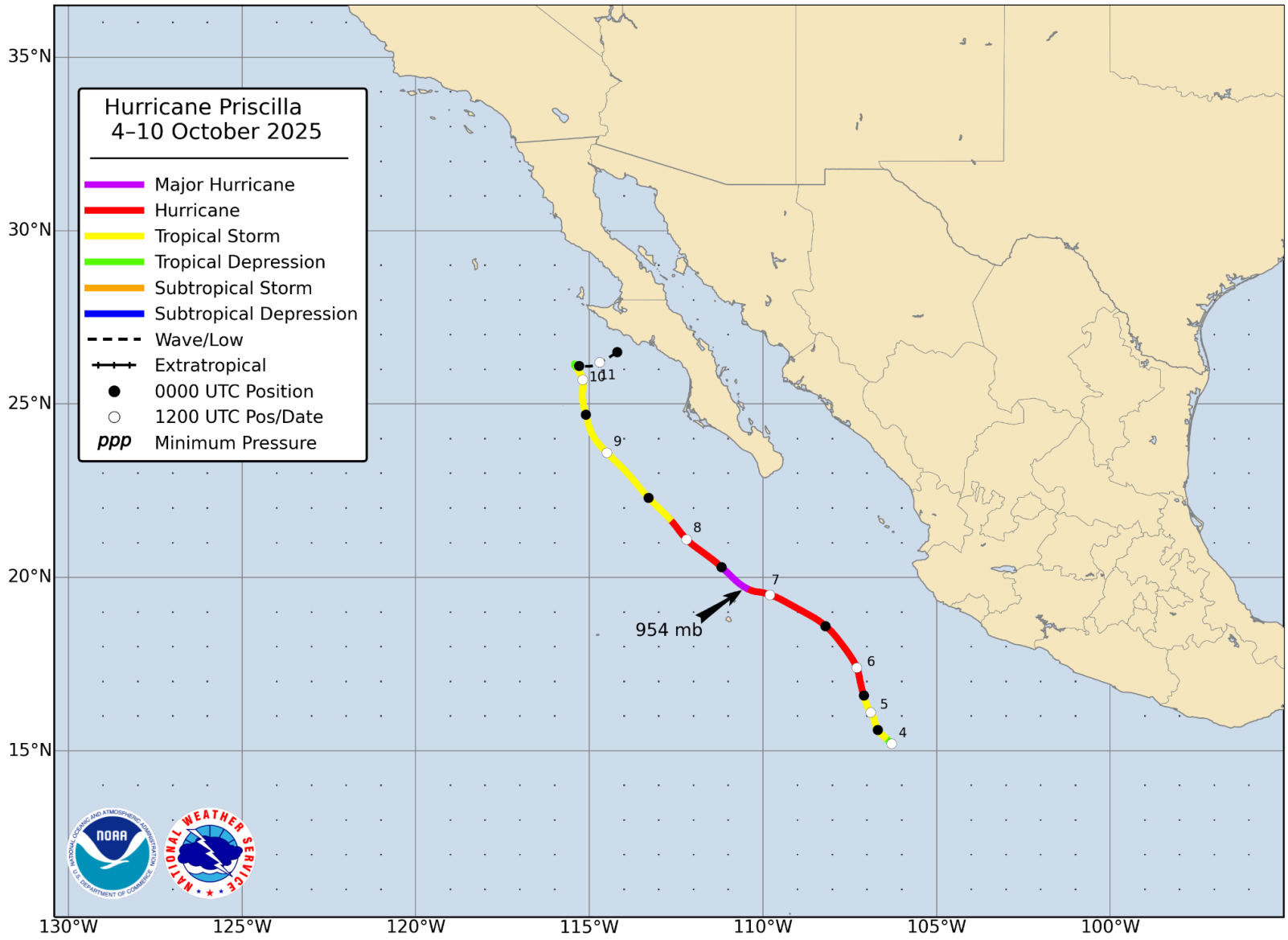


Figure 1. Best track positions for Hurricane Priscilla, 4–10 October 2025.

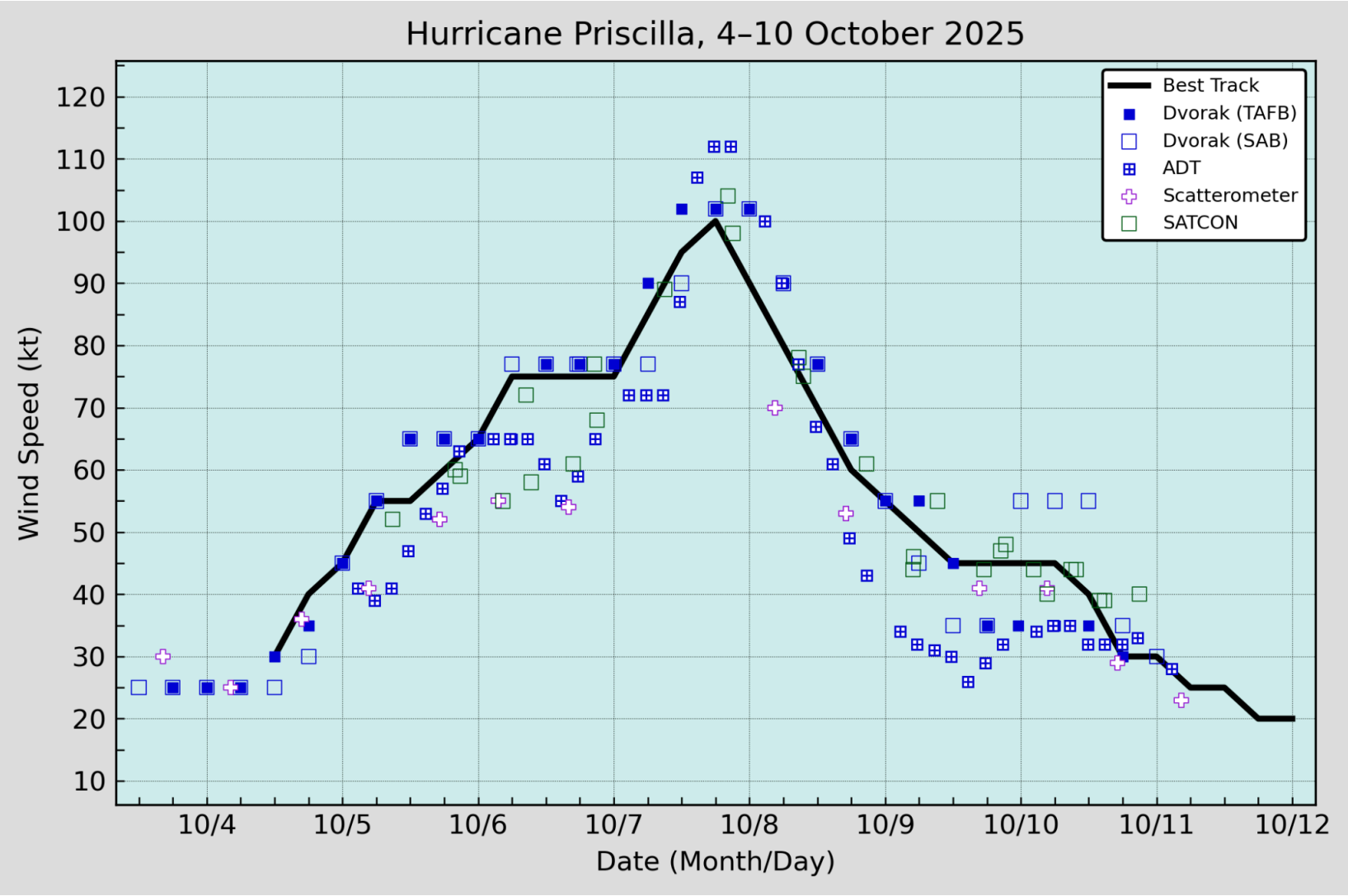


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Priscilla, 4–10 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

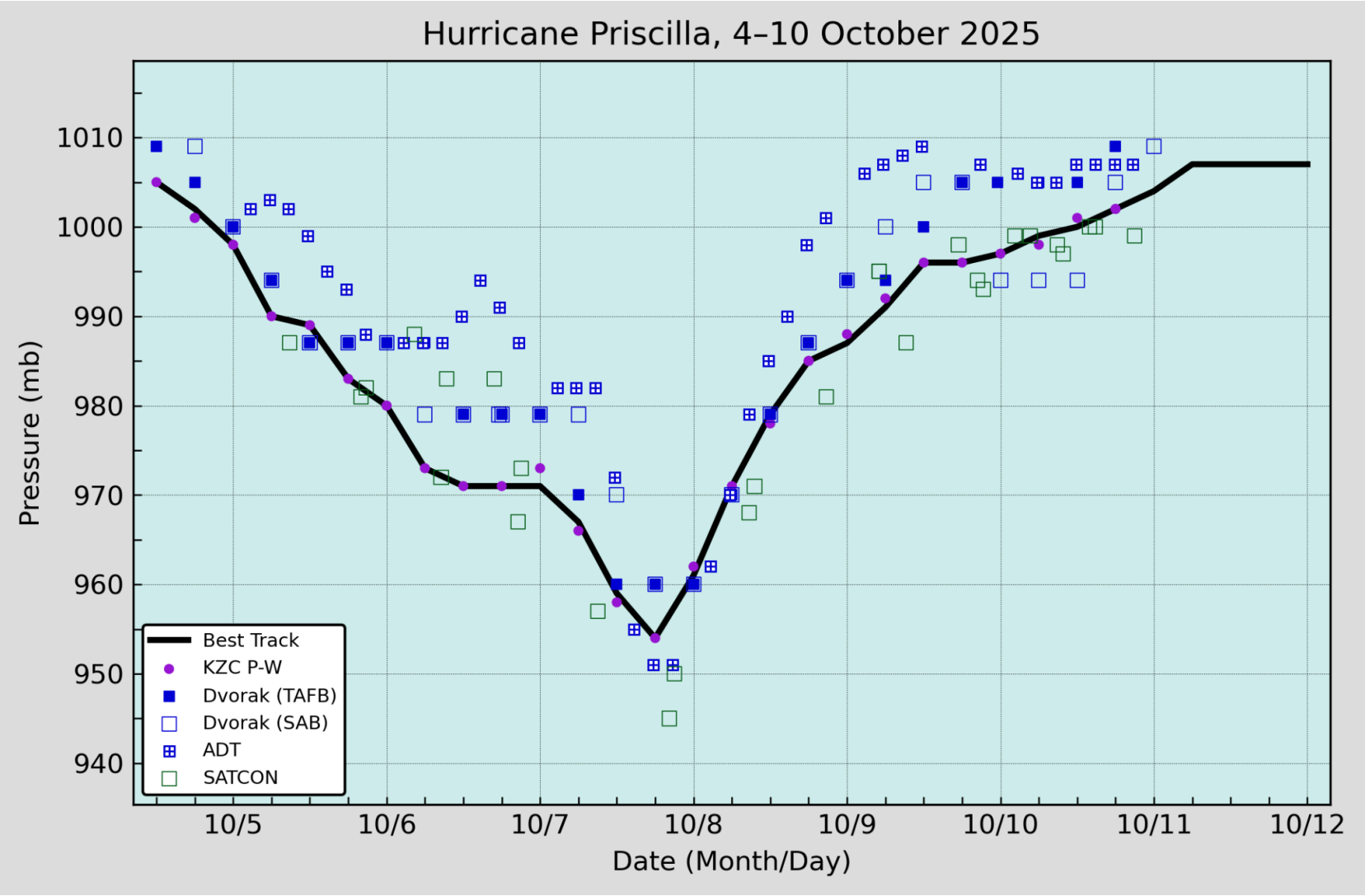


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Priscilla, 4–10 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

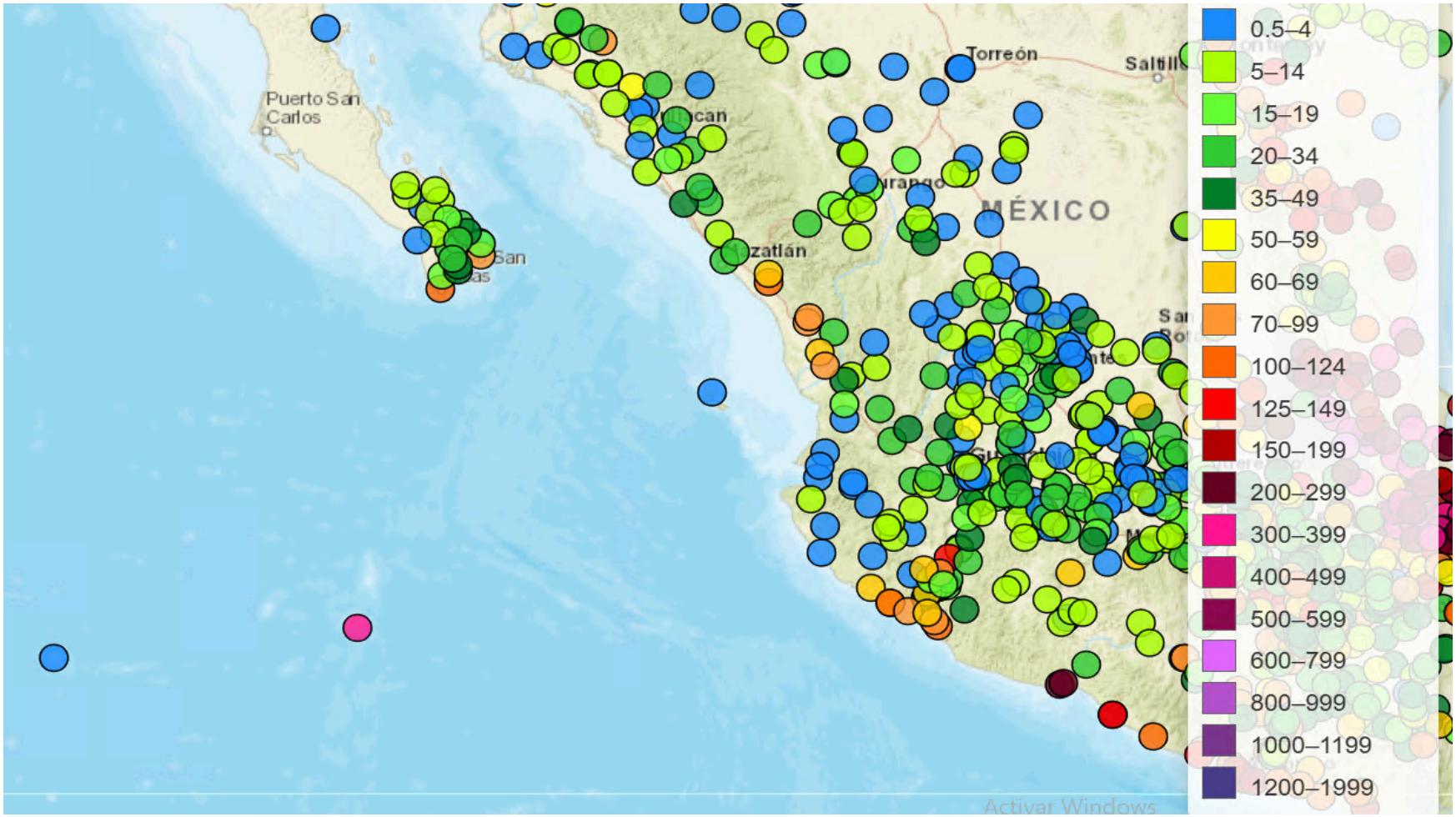


Figure 4. Rainfall accumulations (mm) at stations in Mexico from 4–10 October 2025. Image courtesy of CONAGUA and Servicio Meteorológico Nacional (SMN, the National Meteorological Service of Mexico).

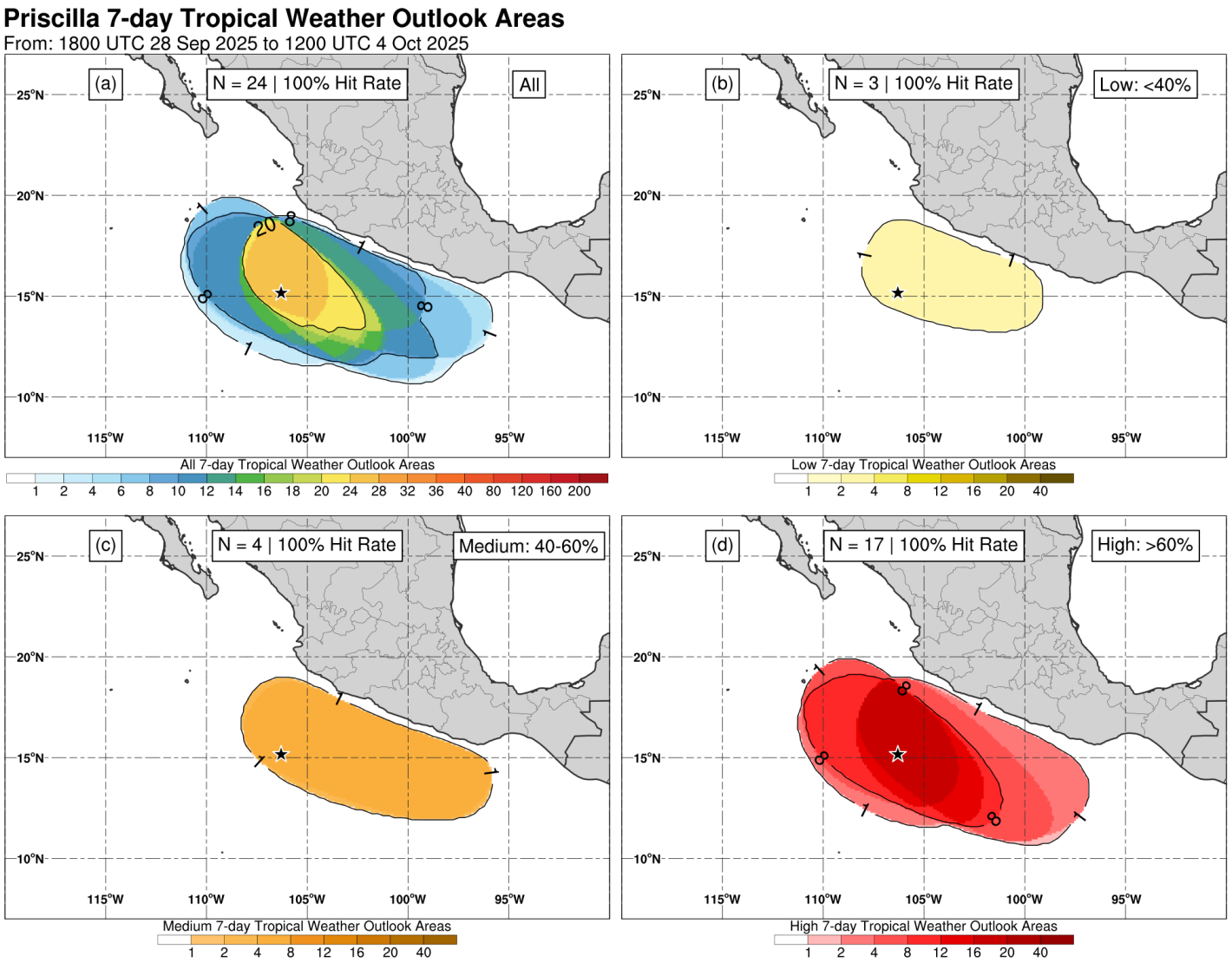


Figure 5. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Priscilla for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.